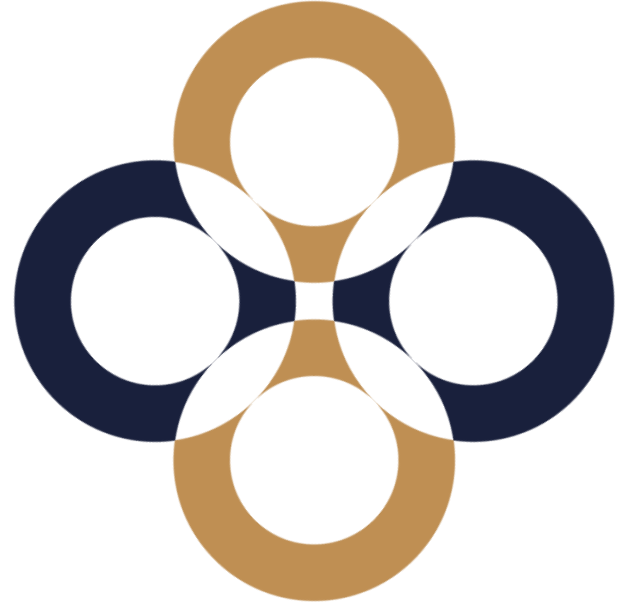


Multiple-based valuation methods

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Multiple-based methods



Comparable
transactions



Market
multiples

Comparable transactions

Based on **earlier actual take-over transactions**

Distortions

The markets change over time
Synergic premiums



Based on **current stock exchange quotes**



DOW J	15,075.18	- 43.31
NASDAQ	3,436.63	+ 0.05
AAPL	456.95	+ 3.98
S&P 500	1,632.57	- 1.13
GOOG	878.04	- 2.19
FTSE 100	6,631.76	+ 6.78

Apple Inc.	
Open: 451.53	Mkt Cap: 428.3B
High: 457.90	52w High: 705.07
Low: 451.50	52w Low: 385.10
Vol: 7.544M	Avg Vol: 17.94M
P/E: 10.89	Yield: 2.70%

Live Quote

Distortions

Market efficiency
Minority discount



Are all watches similar?



What is the value of a book store with a sales of 200 M HUF if another book shop of 150 M HUF revenue has recently been sold for 160 M HUF?

$$P_1/S_1=160/150=1.07$$

$$P_2=1.07*200=\underline{213.3 \text{ M HUF}}$$

Pros and Cons



**Quick
Simple
Reflects the current
market conditions**

**Implicit assumptions
Quality of data
Comparability
Bubbles
Any value could be
justified**



The multiple dictionary

In multiples	In valuation
P	Price of one single share
E (EPS)	After tax profit/number of shares
$P/E = \text{Price} / \text{EPS}$	$E / \text{Profit after tax}$
EV (Enterprise value)	$E + D - \text{Cash}$
EV/BV (Book Value)	$(V - \text{Cash}) / \text{IC}$
P/BV	$E / \text{Book value of equity}$
$P/S = \text{Price} / \text{Sales per share}$	E / Sales
EV/S	$(V - \text{Cash}) / \text{Sales}$
CF	Operative CF (usually)

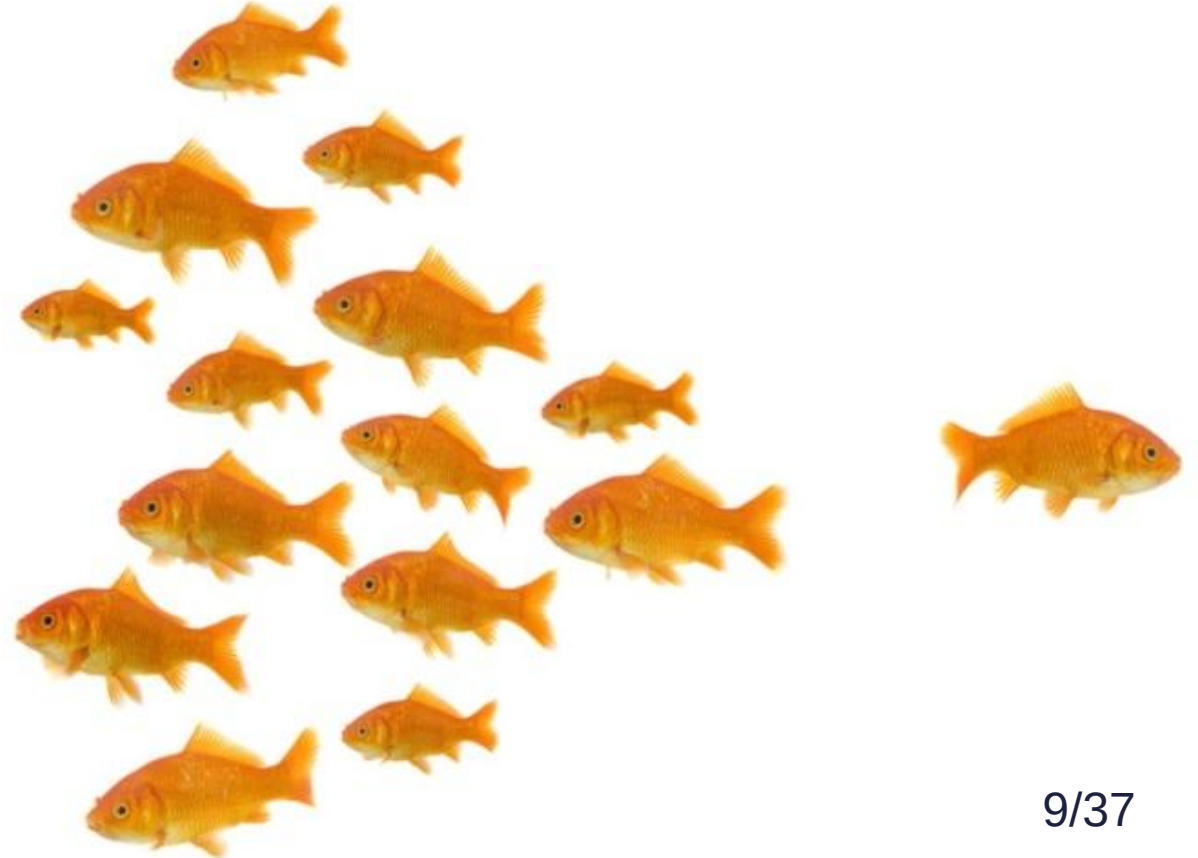
Grouping the multiples

**Profit-
based**

**Revenue-
based**

**Asset-
based**

**Industry
specific**



Defining a multiple

Simple

Easy-to-
understand

Clear

Properly
defined (last,
average,
expected,
extraordinary)

Accessible

We have the
data needed

Choosing the right ratio

Relevance

Either the numerator or the denominator should be some kind of value.

Consistency

The numerator and the denominator should be of the same dimension (V or E).

Uniformity

Same financial year, accounting standard, currency

Empirical tests for applicability

Description power:

What sample size is needed?

What is a high/low value?

Distribution of historical multiple value

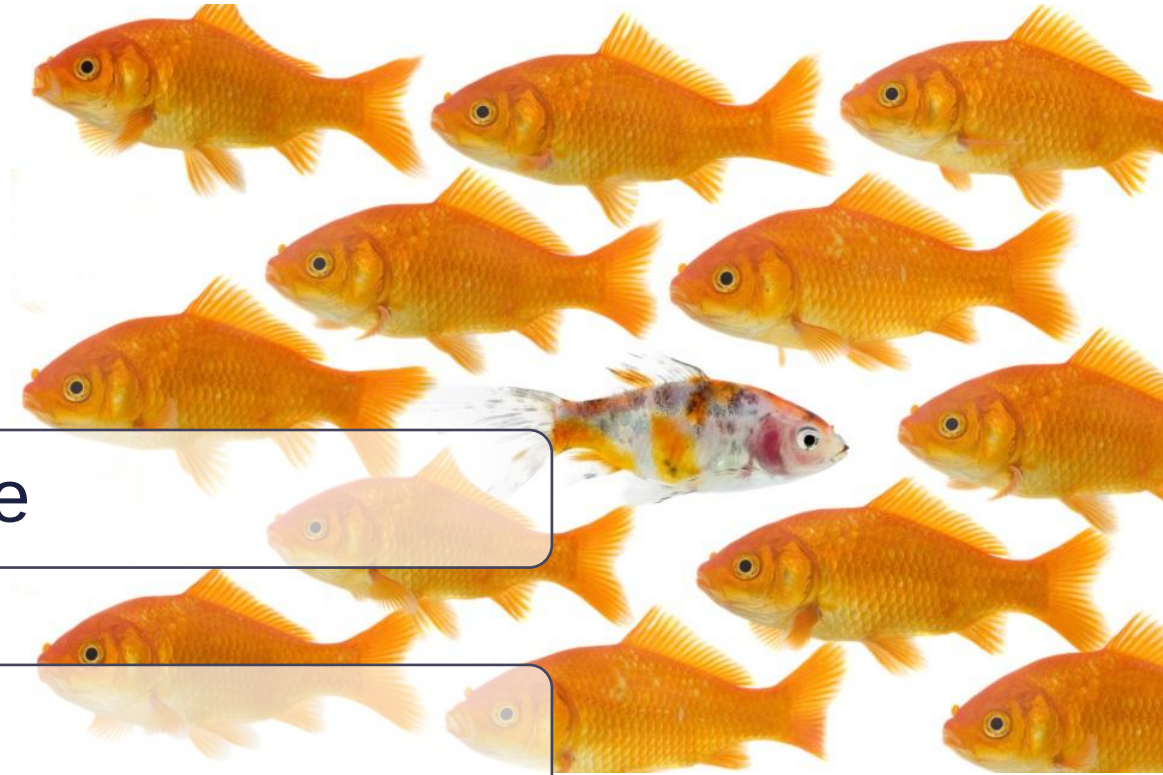
Is the population homogeneous?

Outliers, averages

What needs to be deleted? What would the average show?

Calculation challenges

Negative profit or equity → distortions



Sub-sector multiples

CONSTRUCTION INDUSTRY	EBITDA (M USD)	EV/EBITDA MEAN	25TH-90TH PERCENTILE RANGE	MEDIAN REVENUE (M USD)
Electrical Construction	\$0.3 – \$3.0	4.6x	+44% 2.8-7.4x	\$9.6
Commercial & Heavy Construction	\$0.4 – \$2.8	4.1x	3.0-10.4x	\$9.4
Misc Construction Trades	\$0.4 – \$8.0	4.7x	2.8x-7.8x	\$7.7
Plumbing, HVAC	\$0.4 – \$4.0	4.4x	3.5x-7.9x	\$7.3
Residential Home Construction	\$0.3 – \$1.3	3.2x	2.8x-4.1x	\$6.6
All Construction	\$0.3 – \$8.0	4.5x	2.9x-8.2x	\$8.0

Comparable transactions, US, 2017-2023

Source: <https://www.bmimergers.com/construction-industry-multiples/>

Software companies, 2024

M&A Transactions: Size effect on multiples

Deal size, m USD	Number of deals	Median EV/Revenue	Median EV/EBITDA
<5	186	1.4x	14.5x
5-20	478	2.1x	10.8x
20-50	307	2.4x	12.8x
50-100	251	3.2x	14.8x
100-500	410	4.6x	18.4x
500+	360	5.8x	22.4x
Total	1,992	3.1x	16.4x

Comparable transactions, US

Source: <https://aventis-advisors.com/software-valuation-multiples/>

Software companies, 2024

M&A Transactions: Geography effect on multiples

Region	Number of deals	Median size, m USD	Median EV/Revenue	Median EV/EBITDA
USA	568	223	4.2x	21.6x
United Kingdom	215	34	3.1x	15.5x
China	172	80	4.7x	21.9x
South Korea	79	36	3.1x	19.7x
France	69	47	2.4x	13.8x
Sweden	61	16	3.0x	16.8x
Germany	59	46	2.1x	20.2x
Norway	53	15	4.8x	20.1x
Australia	52	34	3.2x	13.1x
India	48	34	4.5x	15.2x
Other	616	27	2.3x	12.7x
Total	1,992	55	3.1x	16.4x

Comparable transactions, US

Source: <https://aventis-advisors.com/software-valuation-multiples/>

What fundamentals influence the value of the multiple?

What are the prerequisites of comparability?

$$P_0 = \frac{DIV_1}{(r_E - g)} = \frac{EPS_1 * div\%}{(r_E - g)}$$

$$\frac{P_0}{EPS_1} = \frac{P}{E} = \frac{div\%}{r_E - g}$$

Dividend pay-off ratio

Growth outlook

Risk of equity

$$\frac{EV}{EBIT} = \frac{E_0 + D_0 - Cash_0}{EBIT_1} = \frac{(1 - t) * (1 - K)}{WACC - g}$$

$$\frac{P}{S} = \frac{P_0}{Sales_1} = \frac{\frac{EPS}{Sales} * div\%}{r_E - g}$$

$$\frac{EV}{Sales} = \frac{E_0 + D_0 - Cash_0}{Sales_1} = \frac{\frac{EBIT}{Sales} * (1 - t) * (1 - K)}{WACC - g}$$

$$\frac{P}{BV} = \frac{P_0}{BV(E)_0} = \frac{ROE * div\%}{r_E - g}$$

$$\frac{EV}{BV} = \frac{EV_0}{BV(E + D)_0} = \frac{ROI * (1 - t) * (1 - K)}{WACC - g}$$

The process of multiple-based valuation

0. Pick a multiple.

1. Choose comparable firms. Similar risk, growth, profit generation (No industry or size requirement theoretically.)

2. Control for differences. (Similar but not the same.)

Subjective corrections (weighted average)

Modified multiples (corrected for differences)

3. Check reasonability.

Industry regressions

Market regressions

Assumptions of multiple types 1

Profit-based (P/E , $EV/EBIT$): the same link between the right CF and the given profit.
(g, CF-generation, risk)

Revenue-based (EV/S): from a given amount of revenue any firms generates the same profit.
(g, CF-generation, risk, „margin”: CF/S)



Assumptions of multiple types 2

Asset-based (P/BV, EV/IC): the assets generate the same profit everywhere.

(g, CF-generation, risk, ROI(c)/ROE)

Industry-specific (EV/employee): the given specific variable creates the same profit everywhere
(g, CF-generation, risk, „industry”: same technology and efficiency)



P/E effects of differences

$$\frac{P}{E} = \frac{div\%}{r_E - g}$$

Interest rates $\uparrow \rightarrow$ Cost of capital $\uparrow$, P/E $\downarrow$

Risk taking willingness $\uparrow \rightarrow r_E \downarrow$, P/E $\uparrow$

General market outlook $\rightarrow$ average $g \uparrow$, P/E $\uparrow$

Profitability $\uparrow \rightarrow div\% \uparrow$, P/E $\uparrow$

Do not forget that in the long run

$$ROE * (1-div\%) = g^*$$

High P/E:

High dividend pay-off

Low r-g

- r big & g huge
- r small & g tiny

P/E/g (PEG)

Relative P/E = $(P/E_{\text{Firm}})/(P/E_{\text{Industry}})$



		PEG	
		Low	High
P/E	Low	r high, g low	r high, g very low
	High	r high, g high	r low, g low

Divided payoff ratio?

Only true for firms with the same dividend policy.

Revenue-based multiples

Easy to use
Always positive
No accounting
issues

Very far from the
CF

P/S is inconsistent.

$$\frac{EV}{S} = \frac{\frac{EBIT}{Sales} * (1 - t) * (1 - K)}{WACC - g}$$



Stable, easy-to-compare,
loss making firms create no issue



Accounting policy differences,
off-balance sheet items, $BV(E) < 0$

$P / BV(E)$

$$P_0/BV_0 = ROE * (1+g_n) * OFH / (r-g_n) = ROE * P/E$$

$$\underline{EV / BV (D+E) (MV/BV)}$$

$$= ROIC * (1-K) / (WACC - g)$$

$$ROIC = NOPLAT / BV(D+E)$$

$$\text{Stability: } ROIC * K = g \rightarrow$$

$$K = g / ROIC$$

$$EV/BV = (ROIC - g) / (WACC - g)$$



James Tobin
(1918-2002)

Asset-based multiples 3

Tobin's Q

= Market value of Assets / Replacement value of Assets

Original use: When shall we purchase a stock?

NPV (added value) ratio

Importance of off-balance sheet items

Hard to use for valuation (no E or EV in the formula)

Value creation linked to operation

No accounting issues

No further information needed

Same technology and efficiency?

Examples

Number of subscribers, readers, visitors, buyers, patients, quantity produced, owned resource confirmed, sales area, number of rooms

Special sector multiples

EV/EBITDAR	Retail, airlines	Without rental fees and lease
EV/EBITDAX	Oil and gas, mining	Without exploration expenses
EV/Resource	Oil and gas, mining, lumber	
EV/Capacity	Oil and gas refinery, hotels, restaurants	
EV/Production (Service)	Oil and gas, harbours, airports	
EV/Subscriber	Telecom, media, internet services	
P/tBV	Banks	Tangible book value

A real-life example 1

BANKS

	Price (local currency)	2013	P/E 2014F	2015F	P/tBV 2014F	2015F	ROE 2014F	Div. yield 2014F
OTP	4,285	18.7	-11.5	11.1	1.0	1.0	-7.3	3.9
Erste Bank	23.4	158.3	-6.7	12.9	1.2	1.1	-14.2	0.0
FHB	797.0	n.a.	n.a.	95.9	3.1	3.1	-22.2	0.0
PKO BP	32.6	12.6	12.1	11.6	1.6	1.5	12.7	4.5
Pekao	185.55	17.5	16.7	15.5	2.1	2.1	12.4	5.4
Raiffeisen	13.5	4.7	-10.6	8.7	0.5	0.4	-3.9	0.0
Bank Handlowy	106	14.6	15.8	14.3	n.a.	n.a.	11.8	6.2
Komerční Banka	5,304	15.5	15.7	14.9	1.8	1.6	12.4	5.6
Unicredit	5.95	16.1	12.4	10.0	0.7	0.7	5.3	2.7
Intesa	2.98	36.9	16.1	13.0	n.a.	n.a.	6.5	3.8
Peer group's average		34.5	11.2	20.8	1.5	1.4	1.3	3.2
Peer group's average excl. FHB		34.5	11.2	12.4	1.3	1.2	4.0	3.6

Source: Bloomberg, Concorde estimates (OTP, Erste Bank, FHB, PKO BP, Pekao, Raiffeisen)

Source: Concorde, 2 March 2015

A real-life example 2

UTILITY

	Price (local curr.)	2013	P/E 2014F	2015F	P/BV 2014F	EV/EBITDA 2014F	2015F	Div yield 2014F
CEZ	621.1	9.2	11.2	16.9	1.2	7.2	8.6	5.2
E.ON	14.3	n.a.	16.3	16.5	0.9	5.4	5.7	3.5
RWE	25.0	n.a.	11.5	11.8	2.2	4.5	4.5	3.9
Enel	4.1	13.4	12.9	12.8	1.1	7.0	7.1	3.2
Endesa	18.1	64.9	18.3	17.4	2.2	7.8	7.7	4.9
Iberdrola	6.1	16.5	16.2	15.2	1.1	8.9	8.6	4.5
EdP	3.5	15.9	13.7	14.0	1.5	9.5	9.5	5.4
EdF	24.7	12.4	11.9	12.2	1.3	5.0	5.0	5.1
PGE	20.7	10.6	11.8	12.3	0.9	5.2	5.1	4.3
Tauron	4.9	7.5	7.2	8.3	0.5	4.1	4.2	3.6
Enea	16.2	8.1	8.3	10.6	0.6	4.3	4.5	3.7
Peer group's average		17.6	12.7	13.5	1.2	6.3	6.4	4.3

Source: Bloomberg, Concorde estimates (CEZ)

Source: Concorde, 2 March 2015

A real-life example 3

SOTP VALUATION

	EV/Share [HUF]	EV [HUF million]	Multiple	EBITDA [HUF million]	Margin Forecast
Hungarian Fix	257	267,504	4.7x	56,916	33%
Hungarian Mobile	549	572,508	5.5x	104,092	36%
SI/IT	127	132,030	9.0x	14,670	22%
Macedonia	69	71,915	3.8x	18,925	38%
EV	1,001	1,043,957	5.8x	194,603	32%
Net debt	-297	-309,976			
Minority	-47	-49,317			
Dividend	-25	1,043			
Spectrum CAPEX	-91	-95,000			
Fair value	540	590,707			
Upside w dividend	27%				

REGIONAL PEERS

	EV/EBITDA	EBITDA margin
Hrvatski Telekom	3.8x	39%
DIGI	4.0x	31%
Telekom Slovenije	3.8x	29%
Telekom Austria	5.5x	32%
Hellenic Telecom	4.6x	32%
Orange Polska	4.6x	25%
Netia	6.3x	27%
Cyfrowy Polsat	6.9x	37%
Megafon	4.1x	32%
Rostelecom	3.6x	31%
Turk Telekom	4.8x	34%
AVG	4.7x	32%

Source: Bloomberg

32/37

Source: Concorde, 2 March 2018

What if not all similarity requirements are fulfilled?

Recalculated multiples

Our firm differs from the average (g , r , CF , β).
Is that maintainable in the long run?

Discounts & premiums

The peers are different in a given perspective
(other markets, countries, liquidity).

Recalculated multiple

The average forward P/E of European food wholesalers was **15.2** at the beginning of 2019. (Div%=93.07%, g=7.30%*) The firm to value has the same risk, but div%=80%, g=8%.

$$\text{div}\% / \text{P/E} = r - g = 93.07\% / 15.20 = 6.12\%$$

$$r - g_{\text{mod}} = 6.12\% - (8.00\% - 7.30\%) = 5.42\%$$

$$\text{P/E}_{\text{mod}} = 80.00\% / 5.42\% = \mathbf{14.76}$$



*Source: Damodaran Online, 2019

The operator of a 20-room hotel with unique regional traits is to value. Similar quality hotels are quoted at the stock exchange at 8 M HUF/room.

$$20 * 8 = 160 \text{ million HUF}$$

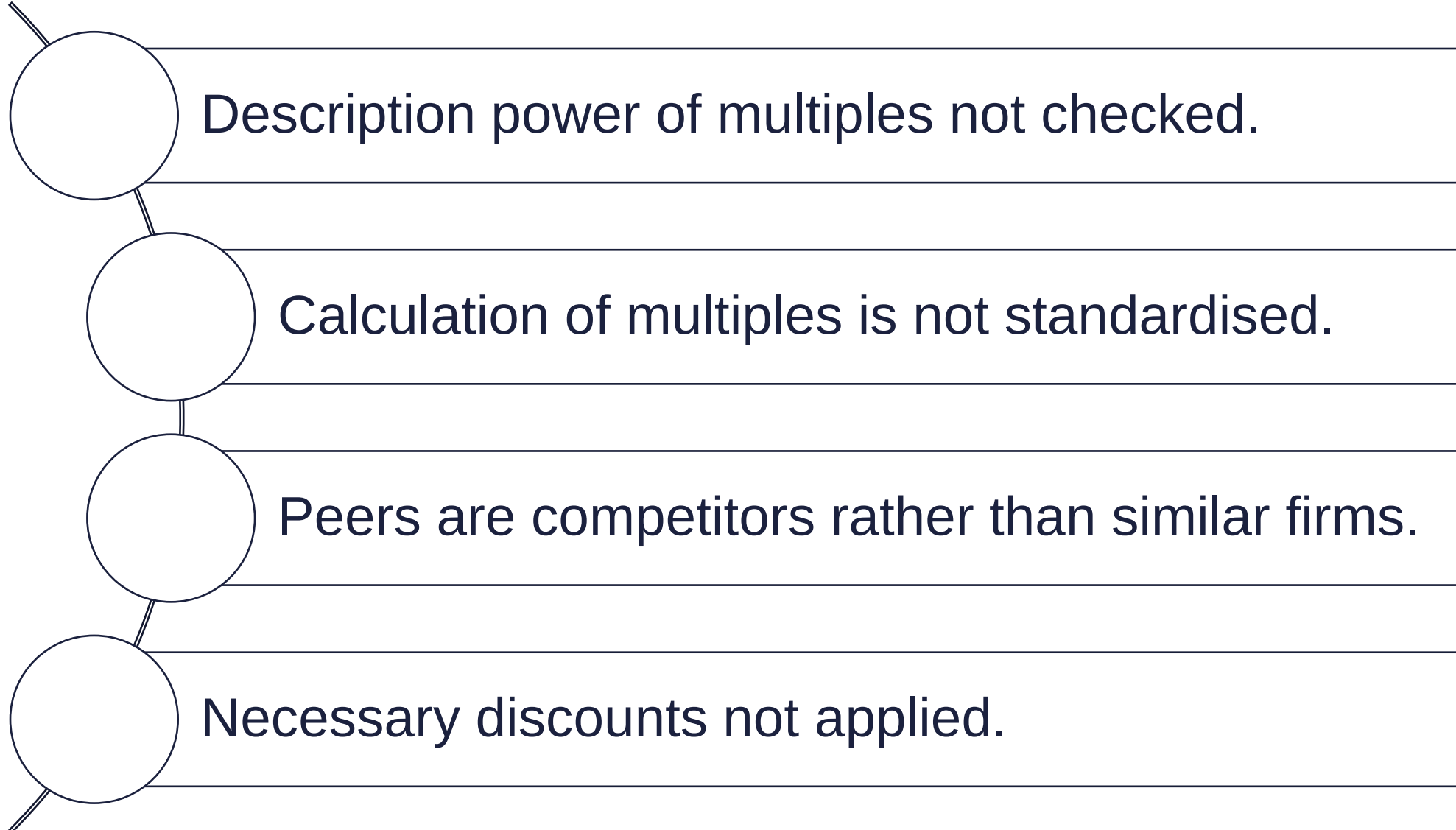
Long-run competitive advantage: + 10%

Illiquidity of the investment: – 30%

$$160 * 1.1 * 0.7 = \underline{123.2 \text{ million HUF}}$$



Typical mistakes when using multiples



**Thank you
for your attention.
Any questions?**

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